

Curriculum Vitae
Melissa N. Horger, PhD

EDUCATION

2016-2021 **Doctor of Philosophy**, Department of Psychology
Program in Developmental Psychology
Graduate Center, City University of New York
Advisor: Sarah Berger, Ph.D.

2019 **Master of Arts**, Department of Psychology
Program in Developmental Psychology
Graduate Center, City University of New York
En-Route Master degrees

2013 **Bachelor of Arts**, Department of Psychology
Millersville University
Honors College
Millersville, PA 17551

ADDITIONAL TRAINING

July 2023 Training in MR Methods, Dr. Tracy Riggins and the
Neurocognitive Development Lab, University of Maryland

March 2023 Young Investigators Research Forum, American Academy of
Sleep Medicine

March 2021 Modern Longitudinal Data Analysis, Dr. Jeffery Harring, University
of Maryland

May 2019 Longitudinal Structural Equation Modelling, Curran-Bauer
Analytics

July 2017 Methodological training on actigraphy from Dr. Anat Scher at the
University of Haifa

RESEARCH EXPERIENCE

July 2023-Present **Postdoctoral Fellow**, Department of Psychology and Brain
Sciences, University of Massachusetts Amherst, Amherst, MA

Aug 2021-July 2023 **Postdoctoral Researcher**, Department of Psychology and
Neuroscience Temple University Philadelphia, PA

Aug 2016-Aug 2021 **Graduate Research Assistant**, Child Development Lab, CUNY
New York, NY

MENTORSHIP EXPERIENCE

July 2023-Present Somneuro Lab: 24 students under my direct supervision

- 1 undergraduate honors thesis
- 1 received funding
 - UMass Honors Research Grant

Aug 2021-July 2023 Developmental Science Lab: 15 undergraduate students under my
direct supervision

- 1 undergraduate honors thesis
- 1 received funding
 - Diamond Scholars Research Stipend

Aug 2016-Aug 2021 Child Development Lab: 26 students under my direct supervision

- 5 undergraduate honors theses
- 2 received funding for their research

- College of Staten Island Technology Grant and Research Scholarship
- CUNY Coordinated Undergraduate Education (CUE) Undergraduate Research Stipend; ICIS Undergraduate Summer Research Fellowship

TEACHING EXPERIENCE

- Spring 2023 **Adjunct Instructor**, Department of Psychology and Neuroscience, Temple University
Course taught:
 Phases of Development: Infancy (Psyc 3301) 3 credits
- Fall 2018-Fall 2020 **Adjunct Instructor**, Department of Psychology College of Staten Island, CUNY
Courses taught:
 Experimental Psychology: Child Development (Psyc 335) 6 credits
 Introduction to Psychology (Psyc 100) 3 credits

ADDITIONAL EXPERIENCE

- 2014-2016 **Educational Behavioral Support, Behavior Technician**
 The Vista School
 Hershey, PA 17003
- 2013-2015 **Therapeutic Support Staff**
 Community Services Group
 Lancaster, PA 17603

PUBLICATIONS

- Campbell, K., **Horger, M. N.**, Pavlakos, G., Simonetti, V., Bulgheroni, M., Parma, V., & Marshall, P. J. (under review). Computational insights into infant prehension: A longitudinal, naturalistic approach. *Developmental Science*.
- Gaudette, L.*, Swift, A.*, **Horger, M. N.**, Holmes, J., & Spencer, R. M. C. (under review). Pediatric sleep electrophysiology: Using polysomnography in developmental cognitive neuroscience. *Developmental Cognitive Neuroscience*.
- Riggins, T., Ratliff, E. L., **Horger, M. N.**, & Spencer, R. M. C. (2024) The importance of sleep for the developing brain. *Current Sleep Medicine Reviews*, 1-10.
- Horger, M. N.**, Campbell, K., & Marshall, P. (2024) Electrophysiological responses to digit stimulation in a tactile oddball paradigm. *International Journal of Psychophysiology*, 203, 112391.
- Horger, M. N.**, DeMasi, A., Allia, A. M., Scher, A., & Berger, S. E. (2023). The unique contributions of day and night sleep on infant motor problem solving. *Journal of Experimental Child Psychology*, 226, 105536.
- DeMasi, A., **Horger, M. N.**, Scher, A., & Berger, S. E. (2022). Infant motor development predicts the dynamics of movement during sleep. *Infancy*, 1– 21.
- Horger, M.N.** (2022). Novel and noninvasive methods for in-home sleep measurement and subsequent state coding in 12-month-old infants. *Infant Behavior and Development*, 69, 101775.
- DeMasi, A., **Horger, M. N.**, Allia, A. M., Scher, A., & Berger, S. E. (2021). Nap timing makes a difference: Sleeping sooner rather than later after learning improves infants' locomotor problem solving. *Infant Behavior and Development*, 65, 101652.
- Berger, S. E., Karasik, L. B., DeMasi, A. A. & **Horger, M. N.** (2021). *Motor development: New perspectives*. Oxford Research Encyclopedia.

- Horger, M. N.**, Marsiliani, R. S., DeMasi, A., Allia, A., & Berger, S. E. (2021). Researcher choices for infant sleep assessment: Parent report, actigraphy, and a novel video system. *Journal of Genetic Psychology*, 182, 218-235.
- Horger, M. N.**, DeMasi, A., Allia, A., Scher, A., & Berger, S. E. (2021). Newly walking infants' night sleep impacts next day learning and problem solving. In S. E. Berger, A. Scher & R. T. Harbourne (Eds.), *Advances in Child Development and Behavior, Volume 60, Interdisciplinary Perspectives on the Relation between Sleep and Learning in Early Development* (pp. 57-83). San Diego, CA: Elsevier.
- Horger, M. N.** & Berger, S. E. (2019). The role of walking experience on whole-body exploration and problem solving. *Cognitive Development*, 52, 1-10.
- Berger, S. E., Harbourne, R. T., and **Horger, M. N.** (2018). Cognition-action trade-offs reflect the organization of attention in infancy. In J. B. Benson (Ed.), *Advances in Child Development and Behavior, Volume 54* (pp. 45-86). San Diego, CA: Elsevier.
- Rush, K. S., Golden, M. E., Mortenson, B. P., Albohn, D., and **Horger, M.** (2017). The effects of a mindfulness and biofeedback program on the on- and off- task behaviors of students with emotional behavioral disorders. *Contemporary School Psychology*, 18, 347-357.

RESEARCH GRANTS & AWARDS

Spring 2025	F32 (NIH), "Understanding the relationship between sleep, motor learning, and motor development in early childhood.", \$157,151
Spring 2023	AASM Young Investigators Research Forum
Spring 2022	Dissertation Showcase (The Graduate Center, CUNY), \$250
Fall 2020	ISDP Student/Postdoc Abstract Award
Summer 2020	Graduate Research Grant (Psi Chi) "The relationship between learning and sleep states (REM and nonREM) during infancy.", \$1500
Spring 2020	Dissertation Award (The Graduate Center, CUNY), "The relationship between learning and sleep states (REM and nonREM) during infancy.", \$10,000 Doctoral Student Research Grant (The Graduate Center, CUNY), "The relationship between learning and sleep states (REM and nonREM) during infancy.", \$1400.
Fall 2019	Psychology Travel Award (The Graduate Center, CUNY), \$500
Summer 2019	Provost Pre-Dissertation Fellowship (The Graduate Center CUNY), "An investigation of the relationship between REM and nonREM and motor learning.", \$4000
Spring 2018	Doctoral Student Research Grant (The Graduate Center CUNY), "The use of noninvasive technology to code sleep states in infants", \$1500
Spring 2017	Conference Presentation Support (The Graduate Center CUNY), \$300
Spring 2013	Psychology Department Senior Award, Richard J Hess Memorial Scholarship in Psychology Susan Luek Graduate Study Award in Psychology (Millersville University)
Spring 2012	Neimeyer-Hodgson Grant Class of 1948 Endowment Psychology Department Junior Award (Millersville University)
Spring 2011	Student Research Grant (Millersville University)

CONFERENCE PRESENTATIONS (*indicates a mentored undergraduate student)

- Horger, M. N.**, Swift, A. M., & Spencer, R. M. C. (2025, June). *Frontal and Central Spindle Density and Developmental Outcomes in Infancy*. Abstracted accepted to be presented as a poster and oral presentation at SLEEP, Seattle, WA.
- Swift, A. M., **Horger, M.**, Holmes, J., Riggins, T., & Spencer, R. M. C. (2025, June). Longitudinal Study of Sleep Duration and Variability Across Early Childhood. Abstract accepted to be presented as a poster at SLEEP, Seattle, WA.
- St. Laurent, C. W., **Horger, M. N.**, Mooney, L., Riggins, T., & Spencer, R. M. C. (2024, September). *Influences of Physical Activity and Sleep on Brain Volume in Early Childhood*. Poster presented at Flux, Baltimore, MD.
- Swift, A. M., **Horger, M. N.**, Holmes, J., Riggins, T., & Spencer, R. M. C. (2024, September). *Slow Wave Activity and Declarative Memory Across the Transition from Triphasic to Biphasic Sleep*. Poster presented at Flux, Baltimore, MD.
- Kohn, B.H., Ratliff, E.L., **Horger, M. N.**, Riggins, T. (2024, September). *Adolescent, but Not Early-Life, Caregiver Depression Is Associated with Smaller Hippocampal Volumes in Adolescents with a History of Prenatal Drug Exposure*. Poster presented at Flux, Baltimore, MD.
- Mooney, L., **Horger, M. N.**, Ratliff, E., St Laurent, C., Riggins, T., & Spencer, R. M. C. *Individual Differences in Hippocampal Volume in Relation to Sleep Morphology*. Poster presented at Flux, Baltimore, MD.
- Horger, M. N.**, Holmes, J. F., Swift, A. M. & Spencer, R. M. C. (2024, July). *The Impact of Sleep Spindles on Fine and Gross Motor Development*. Paper presented at the International Congress of Infant Studies, Glasgow, Scotland.
- Horger, M. N.**, Swift, A. M, Kainec, K., Holmes, J., and Spencer, R. M. C. (2024, June) *Spindle Detection in Polysomnography of Toddlers: Comparing Manual Feature Detection and an Automated Algorithm*. Poster presented at SLEEP, Houston, TX.
- Campbell, K., **Horger M. N.**, Parma, V., & Marshall, P. (2023, March) *Comparing Naturalistic and Lab Measurements: A Study of the Reach-to-Grasp Movement*. Paper presented at the Eastern Psychological Association, Boston, MA.
- Campbell, K., **Horger, M. N.**, Parma, V., & Marshall, P. (2022, November) *Sex Differences in the Maturation of the Grasp in Infancy: A Longitudinal Kinematic Analysis*. Poster and flash talk presented at the International Society of Developmental Psychobiology, San Diego, CA.
- *Kelly, R., *Mochal, L., *Hill, K., Campbell, K., **Horger, M. N.**, Marshall, P., & Parma, V. (2022, June) *Developmental Changes in Infant Reach to Grasp Behavior: A Kinematic Analysis of Caregiver Cellphone Videos*. Poster presented at the Jean Piaget Society conference, Philadelphia, PA.
- Horger, M. N.**, Campbell, K., Marshall, P., & Parma, V. (2022, July) *Early Grasping Trajectories and Later Neural Representations of Infant Fingers*. Paper presented at the International Congress of Infant Studies, Ottawa, Canada.
- DeMasi, A., **Horger, M. N.**, *Allia, A., & Berger, S. E. (2021, April). *Napping Benefits Infants When They Learn a Challenging Task, but Not an Easy One*. Poster presented at the Biennial Meeting of the Society for Research in Child Development (virtual).
- Soker-Elimaliah, S., Maia, M. G., Jancart, K., *Patel, S., **Horger, M. N.**, Harbourne, R. T., & Berger, S. E. (2021, April). *Sit Up Straight and Focus: The Relationship*

- between Infants' Sitting Ability and Focused Attention*. Poster presented at the Biennial Meeting of the Society for Research in Child Development (virtual).
- Horger, M. N.** & Berger, S. E. (2021, April). *Parental Expectations and Cognitions Surrounding Infant Sleep*. Poster presented at the Biennial Meeting of the Society for Research in Child Development (virtual).
- Horger, M. N.** & Berger, S. E. (2021, February). *Noninvasive, In-Home State Coding in Infants and Toddlers Using an Automated Coding System*. Poster presented at the International Pediatric Sleep Association (virtual).
- Horger, M. N.**, *D'Aloia, R. E., *Allia, A. M., *Neumann, T., & Berger, S. E. (2020, October). *Changes in Infant Sleep States Around Motor Milestone Acquisition*. Poster presented at the International Society of Developmental Psychobiology (virtual).
- Horger, M. N.**, DeMasi, A., *Allia, A. A., Marsiliani, R. S. & Berger, S. E. (2020, July). *Task Difficulty Moderates the Impact of Learning on Fragmentation (but Not Duration) during REM-Dominant Sleep*. Poster presented at the International Congress of Infant Studies (virtual).
- DeMasi, A., *Allia, A. A., **Horger, M. N.**, & Berger, S. E. (2020, July). *Prior Locomotor Experience and Sleep Quality Predict Newly Walking Infants' Learning Trajectories*. Poster presented at the International Congress of Infant Studies (virtual).
- DeMasi, A., Berger, S. E., **Horger, M. N.**, & *Allia, A. A. (2020, July). *Movement during Nighttime Wake Episodes Reflects the Onset of Crawling and Walking*. Poster presented at the International Congress of Infant Studies (virtual).
- Berger, S. E., **Horger, M. N.**, DeMasi, A., & Barnett, N. L. (2019, November). *Video from the Crib: Using the Nanit Home Baby Monitoring System to Explore the Relation between Motor Development and Sleep in Infancy*. Poster presented at the 10th Biennial Pediatric Sleep Medicine Meeting, Naples, FL.
- Horger, M. N.**, Berger, S. E., Barnett, N. L. & Glazer, A. (2019, November). *Methods for Infant Sleep Measurement: A Comparison of Parent Report, Actigraphy, and the Nanit Video Monitoring System*. Poster presented at the 10th Biennial Pediatric Sleep Medicine Meeting, Naples, FL.
- Hinav A., **Horger M. N.**, Berger S., & Scher A. (2019, September). *Novice Walkers Perform a Challenging Task: Does Temperament Play a Role?* Poster presented at the 19th European Conference on Developmental Psychology, Athens, GREECE.
- Simon, T., **Horger M. N.**, Berger S. E., & Scher A. (July, 2019). *Novice Walkers Solve a Motor Challenge: Does Sleep Quality Predict Performance?* Poster presented at the 3rd World Conference on Movement and Cognition, Tel Aviv, ISRAEL.
- Horger, M. N.**, Berger, S. E., DeMasi, A., *Allia, A. M., & Scher, A. (2019, June). *Night Sleep and Napping Together Consolidate Infants' Motor Problem Solving*. Poster presented at SLEEP, San Antonio, TX.
- Horger, M. N.**, Marsiliani, R. S., & Berger, S. E. (2019, March). *The Intersection of Motor and Language Skills in Problem Solving*. Paper presented at the Eastern Psychological Association, New York, NY.
- Berger, S. E., Scher, A., & **Horger, M. N.** (2019, February). *Learning to Solve a Novel Locomotor Problem Depends on Quality and Timing of Infants' Sleep*. Poster presented at the 6th Conference on Cognition Research of the Israeli Society for Cognitive Psychology, Akko, ISRAEL.

- Hinav A., **Horger M.**, Berger S., & Scher A. (2018, September). *Novice Walkers Perform a Challenging Task: Does Temperament Play a Role?* Poster presented at the 19th European Conference on Developmental Psychology, Athens, GREECE.
- Horger, M.**, *Saad, P., Scher, A., and Berger, S. (2018, July) *Infants' Ability to Consolidate New Information Depends on the Timing of a Nap Relative to Learning and Test*. Poster presented at the International Congress of Infant Studies, Philadelphia, PA.
- Horger, M.**, *Friedman, D., Scher, A., and Berger, S. (2018, July) *Individual Differences in the Quality of Night Sleep Influence Infants' Preparedness for Learning the Following Day*. Poster presented at the International Congress of Infant Studies, Philadelphia, PA.
- Horger, M.**, *Saad, P., *Zillig, A., and Berger, S. (2017, October) *Regress for Success: Returning to Crawling Places Attentional Demands on New Walkers*. Poster presented at the Cognitive Development Society, Portland, OR
- Horger, M.**, Berger, S., Orr, E., and Benish-Weisman, M. (2017, May) *The Relation between Culture, Parental Values, and Young Children's Play*. Poster presented at the Association for Psychological Science, Boston, MA and Psychology Research Day, New York, NY.
- Horger, M.**, Rush, K., and Gallagher, S. (2013, March) *Reward Allocation Among Collaborating Children* Poster presented at the Eastern Psychological Association, New York, NY
- Sube, C., Lambert, K., vanBuren, T., **Horger, M.**, and Cook, S. (2013, March) *Investigating the Ability of Synesthetes to Process Meaning* Poster presented at the Eastern Psychological Association, New York, NY
- Horger, M.** and Rush, K. (2012, April) *Using Biofeedback Training to Reduce Performance-Related Anxiety in Children* Poster presented at the Hershey Neuroscience Retreat, Hershey, PA
- Horger, M.** and Rush, K. (2012, March) *Using Biofeedback Training to Reduce Performance-Related Anxiety in Children* Poster presented at Millersville University's Student Research Conference, Millersville, PA
- Horger, M.**, Gavin, K., Rush, K., and Mortenson, B. P. (2011, March) *The Reliability and Validity of Forced-Choice Preference Assessments for Infants and Toddlers* Poster presented at the Pennsylvania Applied Behavior Analysis Conference, Philadelphia, PA
- Horger, M.**, Gavin, K., Rush, K., and Mortenson, B. P. (2012, March) *The Reliability and Validity of Forced-Choice Preference Assessments for Infants and Toddlers* Paper presentation at the Eastern Psychological Association, Pittsburgh, PA
- Horger, M.** and Rush, K. (2011, October) *Preference Assessments in Infants and Toddlers* Paper presentation at the PASSHE Psychology Conference. Kutztown, PA
- Horger, M.** and Rush, K. (2011, March) *Social Skills in Children* Paper presentation at the Eastern Psychological Association, Boston, MA

INVITED LECTURES/TALKS

- | | |
|--------------|---|
| January 2023 | <i>Breathing easy and sleeping soundly</i>
Nanit Labs Symposium |
| March 2020 | <i>Sleep as a context for infant learning and development</i>
Engle-Friedman Lab, Baruch College |
| July 2019 | <i>Learning from sleep: Using Nanit to gain insights about infant sleep and motor development.</i> |

Nanit, NYC Development Team
Sleep measurement and development during infancy
University of Haifa, Israel

PROFESSIONAL SERVICE

2020-Present	Independent reviewer for various journals including <i>The Journal of Genetic Psychology</i> , <i>Journal of Sleep Medicine</i> , <i>Infancy</i> , <i>SLEEP</i> , <i>Children</i> , <i>Journal of Experimental Child Psychology</i> .
2020	Alumni mentor (through Millersville University) Student reviewer for the biannual Cognitive Development Society meeting
2019	Student reviewer for the SRCD 2020 Special Topics conference on Learning Through Play and Imagination Mentored reviewer for <i>SLEEP</i>
2018	Student reviewer for the biannual Cognitive Development Society meeting

COMMITTEES

2023-Present	Member of Postdoctoral Research Organizing Social Committee, Orientation Committee, and Professional Development Committee
2018-2019	Student representative for the Curriculum Committee, Developmental Psychology Training Area, Doctoral Program in Psychology, CUNY
2017-2021	Student representative for the Executive Committee, Doctoral Program in Psychology, CUNY
2017-2018	Student representative for the Admissions Committee, Developmental Psychology Training Area, Doctoral Program in Psychology, CUNY
2016-2018	GiGi's Playhouse Inc. Communications Committee

PROFESSIONAL ASSOCIATIONS

American Academy of Sleep Medicine
American Psychological Association
Association for Psychological Science
Cognitive Development Society
Eastern Psychological Association
International Society for Infant Studies
International Society for Developmental Psychobiology
Sleep Research Society
Society for Research in Child Development